

GED Math Answer Key

Number Sense and Operations (30 Questions)

Comprehensive Solutions and Explanations

June 2026

Complete Answer Key with Detailed Solutions

Three Difficulty Levels: Basic, Intermediate, and Advanced

This document provides comprehensive solutions for all 30 GED Math Practice questions, featuring:

- Step-by-step problem-solving strategies
- Mathematical concept explanations
 - Common mistakes to avoid
- Alternative solution methods
 - Real-world applications

Designed to support student learning and understanding of fundamental mathematics skills.

Answer Summary

Q	Answer	Q	Answer	Q	Answer
1	A	11	B	21	B
2	B	12	C	22	B
3	B	13	C	23	C
4	A	14	C	24	D
5	A	15	D	25	D
6	A	16	A	26	C
7	A	17	A	27	D
8	C	18	C	28	C
9	C	19	A	29	B
10	C	20	A	30	B

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1 Basic Level: Fundamental Operations (Questions 1–10)

1.1 Question 1: Order of Operations

Correct Answer: **A**

Problem

Simplify: $8 + 3 \times 5 - 2$

Solution

To solve this problem, we must follow the **Order of Operations**, often remembered by the acronym *PEMDAS* (Parentheses, Exponents, Multiplication/Division, Addition/Subtraction).

1. **Step 1:** Identify the operations: addition, multiplication, and subtraction
2. **Step 2:** Perform multiplication first: $3 \times 5 = 15$
3. **Step 3:** Rewrite the expression: $8 + 15 - 2$
4. **Step 4:** Perform addition and subtraction from left to right:

$$8 + 15 = 23$$

$$23 - 2 = 21$$

Answer: 21

Mathematical Concept

The Order of Operations is a standard convention that ensures everyone solves expressions the same way. Multiplication and division are performed before addition and subtraction (unless parentheses override this).

Common Mistakes to Avoid

- Working strictly from left to right: $8 + 3 = 11$, then $11 \times 5 = 55$, etc. (incorrect)
- Forgetting to multiply before adding: treating it as $(8 + 3) \times 5 - 2$ (incorrect)

Alternative Verification

Using standard notation: $(8) + (3 \times 5) - (2) = 8 + 15 - 2 = 21$

Real-World Application

Order of operations is crucial in budgeting and finance. For example: Starting with \$8, earning \$3 per hour for 5 hours, then spending \$2 gives you: $8 + 3(5) - 2 = \$21$.

1.2 Question 2: Adding Fractions

Correct Answer: B

Problem

Add: $\frac{1}{4} + \frac{1}{3}$

Solution

To add fractions, we need a common denominator.

1. **Step 1:** Find the least common denominator (LCD) of 4 and 3

- Multiples of 4: 4, 8, 12, 16, ...
- Multiples of 3: 3, 6, 9, 12, ...
- LCD = 12

2. **Step 2:** Convert each fraction to have denominator 12:

$$\begin{aligned}\frac{1}{4} &= \frac{1 \times 3}{4 \times 3} = \frac{3}{12} \\ \frac{1}{3} &= \frac{1 \times 4}{3 \times 4} = \frac{4}{12}\end{aligned}$$

3. **Step 3:** Add the fractions:

$$\frac{3}{12} + \frac{4}{12} = \frac{3+4}{12} = \frac{7}{12}$$

Answer: $\frac{7}{12}$

Mathematical Concept

When adding fractions, we can only add the numerators if the denominators are the same. Finding a common denominator allows us to rewrite fractions in equivalent forms.

Common Mistakes to Avoid

- Adding numerators and denominators separately: $\frac{1+1}{4+3} = \frac{2}{7}$ (incorrect)
- Using an incorrect common denominator: Using 7 instead of 12 (incorrect)

Alternative Verification

Convert to decimals: $\frac{1}{4} = 0.25$ and $\frac{1}{3} \approx 0.333$, so $0.25 + 0.333 \approx 0.583$. Check: $\frac{7}{12} \approx 0.583$

Real-World Application

Cooking requires adding fractions: If a recipe calls for $\frac{1}{4}$ cup flour and $\frac{1}{3}$ cup sugar, you need a total of $\frac{7}{12}$ cup of dry ingredients.

—

1.3 Question 3: Subtracting Decimals

Correct Answer: B

Problem

Subtract: $8.92 - 5.45$

Solution

To subtract decimals, align the decimal points and subtract as with whole numbers.

1. **Step 1:** Align the decimal points:

$$\begin{array}{r} 8.92 \\ -5.45 \\ \hline \end{array}$$

2. **Step 2:** Subtract from right to left:

- Hundredths place: $2 - 5$ requires borrowing. $12 - 5 = 7$
- Tenths place: $8 - 1 - 4 = 3$ (after borrowing 1)
- Ones place: $8 - 5 = 3$

3. **Step 3:** Write the result:

$$8.92 - 5.45 = 3.47$$

Answer: 3.47

Mathematical Concept

Decimal subtraction follows the same regrouping rules as whole number subtraction. The decimal point remains in the same position.

Common Mistakes to Avoid

- Misaligning decimal points: $8.92 - 5.45$ calculated as $892 - 545$ (incorrect)
- Forgetting to borrow when necessary, leading to negative intermediate results

Alternative Verification

Addition check: $3.47 + 5.45 = 8.92$

Real-World Application

Shopping and making change: If an item costs \$5.45 and you pay with \$8.92, your change is $8.92 - 5.45 = \$3.47$.

1.4 Question 4: Converting Fractions to Decimals

Correct Answer: A

Problem

Convert to a decimal: $\frac{3}{8}$

Solution

To convert a fraction to a decimal, divide the numerator by the denominator.

1. **Step 1:** Set up the division: $3 \div 8$
2. **Step 2:** Since $3 < 8$, the result is less than 1. Begin with 0.
3. **Step 3:** Perform long division:

$$3 \div 8 = 0 \text{ remainder } 3$$

$$30 \div 8 = 3 \text{ remainder } 6 \quad (\text{first decimal place})$$

$$60 \div 8 = 7 \text{ remainder } 4 \quad (\text{second decimal place})$$

$$40 \div 8 = 5 \text{ remainder } 0 \quad (\text{third decimal place})$$

4. **Step 4:** Combine the results: 0.375

Answer: 0.375

Mathematical Concept

A fraction can be expressed as a division problem. The fraction bar represents division, so $\frac{3}{8} = 3 \div 8$.

Common Mistakes to Avoid

- Computing $8 \div 3$ instead of $3 \div 8$ (incorrect)
- Stopping the division prematurely, giving an approximation instead of the exact value

Alternative Verification

Multiply back: $0.375 \times 8 = 3$

Real-World Application

Engineering and measurements: $\frac{3}{8}$ inch on a ruler is exactly 0.375 inches, important for precision work.

—

1.5 Question 5: Finding Percentages

Correct Answer: A

Problem

15 is what percent of 60?

Solution

Use the percentage formula: $\text{Percentage} = \frac{\text{Part}}{\text{Whole}} \times 100\%$

1. **Step 1:** Identify the part and whole:

- Part = 15
- Whole = 60

2. **Step 2:** Apply the formula:

$$\text{Percentage} = \frac{15}{60} \times 100\%$$

3. **Step 3:** Simplify the fraction:

$$\frac{15}{60} = \frac{1}{4} = 0.25$$

4. **Step 4:** Multiply by 100%:

$$0.25 \times 100\% = 25\%$$

Answer: 25%

Mathematical Concept

A percentage expresses a number as a fraction of 100. The formula relates a part to a whole, giving us the percentage.

Common Mistakes to Avoid

- Reversing part and whole: $\frac{60}{15} \times 100\% = 400\%$ (incorrect)
- Forgetting to multiply by 100 to convert to a percentage

Alternative Verification

Check: 25% of 60 = $0.25 \times 60 = 15$

Real-World Application

Test scores: If you answer 15 questions correctly out of 60 total, you scored 25

—

1.6 Question 6: Multiplying Fractions

Correct Answer: **A**

Problem

Multiply: $\frac{2}{3} \times \frac{1}{5}$

Solution

To multiply fractions, multiply numerators together and denominators together, then simplify.

1. **Step 1:** Multiply numerators: $2 \times 1 = 2$
2. **Step 2:** Multiply denominators: $3 \times 5 = 15$
3. **Step 3:** Write the result:

$$\frac{2}{3} \times \frac{1}{5} = \frac{2 \times 1}{3 \times 5} = \frac{2}{15}$$

4. **Step 4:** Check if simplification is possible:
 - Factors of 2: 1, 2
 - Factors of 15: 1, 3, 5, 15
 - GCD = 1, so the fraction is already in simplest form

Answer: $\frac{2}{15}$

Mathematical Concept

Unlike addition and subtraction, multiplying fractions does not require a common denominator. Multiply straight across and simplify the result.

Common Mistakes to Avoid

- Finding a common denominator (not needed for multiplication)
- Forgetting to simplify the final answer
- Multiplying denominators incorrectly

Alternative Verification

Convert to decimals: $\frac{2}{3} \approx 0.667$ and $\frac{1}{5} = 0.2$, so $0.667 \times 0.2 \approx 0.133$. Check: $\frac{2}{15} \approx 0.133$

Real-World Application

Cooking: If you need $\frac{2}{3}$ of a $\frac{1}{5}$ cup ingredient, you need $\frac{2}{15}$ cup.

1.7 Question 7: Dividing Fractions

Correct Answer: **A**

Problem

Divide: $\frac{2}{3} \div \frac{3}{4}$

Solution

To divide by a fraction, multiply by its reciprocal (the fraction flipped).

1. **Step 1:** Identify the divisor: $\frac{3}{4}$
2. **Step 2:** Find its reciprocal: $\frac{4}{3}$
3. **Step 3:** Change division to multiplication:

$$\frac{2}{3} \div \frac{3}{4} = \frac{2}{3} \times \frac{4}{3}$$

4. **Step 4:** Multiply:

$$\frac{2}{3} \times \frac{4}{3} = \frac{2 \times 4}{3 \times 3} = \frac{8}{9}$$

5. **Step 5:** Simplify (if needed):

- $\text{GCD}(8, 9) = 1$, so $\frac{8}{9}$ is in simplest form

Answer: $\frac{8}{9}$

Mathematical Concept

Dividing by a number is equivalent to multiplying by its reciprocal. For fractions, the reciprocal of $\frac{a}{b}$ is $\frac{b}{a}$.

Common Mistakes to Avoid

- Flipping the wrong fraction: Using $\frac{3}{2}$ instead of $\frac{4}{3}$ (incorrect)
- Performing division instead of multiplication after flipping
- Forgetting the step-by-step process and trying to divide directly

Alternative Verification

Check: $\frac{8}{9} \times \frac{3}{4} = \frac{24}{36} = \frac{2}{3}$

Real-World Application

Splitting portions: If you have $\frac{2}{3}$ of a pizza and want to divide it into $\frac{3}{4}$ -sized slices, you get $\frac{8}{9}$ slices.

1.8 Question 8: Exponents

Correct Answer: C

Problem

Evaluate: 2^5

Solution

An exponent indicates how many times to multiply a base by itself.

1. **Step 1:** Identify the base and exponent:

- Base = 2
- Exponent = 5

2. **Step 2:** Expand the expression:

$$2^5 = 2 \times 2 \times 2 \times 2 \times 2$$

3. **Step 3:** Multiply step by step:

$$2 \times 2 = 4$$

$$4 \times 2 = 8$$

$$8 \times 2 = 16$$

$$16 \times 2 = 32$$

Answer: 32

Mathematical Concept

Exponents are a shorthand for repeated multiplication. The number of times the base is multiplied is determined by the exponent.

Common Mistakes to Avoid

- Multiplying base times exponent: $2 \times 5 = 10$ (incorrect)
- Miscounting the number of multiplications
- Confusing 2^5 with 5^2

Alternative Verification

Powers of 2 sequence: $2^1 = 2$, $2^2 = 4$, $2^3 = 8$, $2^4 = 16$, $2^5 = 32$

Real-World Application

Computing: Binary systems use powers of 2. $2^5 = 32$ represents the number of possible combinations with 5 binary digits.

1.9 Question 9: Square Roots

Correct Answer: C

Problem

Find: $\sqrt{64}$

Solution

The square root of a number is the value that, when multiplied by itself, equals that number.

1. **Step 1:** Ask: What number times itself equals 64?
2. **Step 2:** Test values:
 - $7 \times 7 = 49$ (too small)
 - $8 \times 8 = 64$ (correct!)
 - $9 \times 9 = 81$ (too large)
3. **Step 3:** Verify: $8 \times 8 = 64$

Answer: 8

Mathematical Concept

A square root is the inverse of squaring. If $n^2 = 64$, then $\sqrt{64} = n$. The symbol $\sqrt{}$ represents the principal (positive) square root.

Common Mistakes to Avoid

- Finding half the number: $\sqrt{64} \neq 32$ (incorrect)
- Confusing with division by 2
- Forgetting that both $+8$ and -8 square to 64, but $\sqrt{64}$ refers to the positive root

Alternative Verification

Check: $8^2 = 8 \times 8 = 64$

Real-World Application

Geometry: If a square room has an area of 64 square feet, each side is $\sqrt{64} = 8$ feet.

—

1.10 Question 10: Absolute Value

Correct Answer: C

Problem

Evaluate: $|-5|$

Solution

The absolute value of a number is its distance from zero on the number line, always positive.

1. **Step 1:** Identify the number: -5
2. **Step 2:** Find its distance from 0:
 - The number -5 is 5 units to the left of 0
 - Distance is always positive
3. **Step 3:** Apply the absolute value:

$$|-5| = 5$$

Answer: 5

Mathematical Concept

Absolute value $|x|$ gives the magnitude of a number without regard to its sign. For any real number n :

$$|n| = \begin{cases} n & \text{if } n \geq 0 \\ -n & \text{if } n < 0 \end{cases}$$

Common Mistakes to Avoid

- Keeping the negative sign: $|-5| \neq -5$ (incorrect)
- Confusing absolute value with negation
- Not recognizing that $|5| = |-5| = 5$

Alternative Verification

Both 5 and -5 are distance 5 from zero: $|5| = 5$ and $|-5| = 5$

Real-World Application

Temperature change: A change of -5°F (decrease) has an absolute value of 5°F in magnitude.

2 Intermediate Level: Multi-Step Operations (Questions 11–20)

2.1 Question 11: Complex Order of Operations

Correct Answer: **B**

Problem

Simplify: $24 \div (4 + 2) - 3 + 1$

Solution

Follow PEMDAS, paying special attention to the parentheses.

1. **Step 1:** Parentheses first: $4 + 2 = 6$
2. **Step 2:** Rewrite: $24 \div 6 - 3 + 1$
3. **Step 3:** Division: $24 \div 6 = 4$
4. **Step 4:** Rewrite: $4 - 3 + 1$
5. **Step 5:** Subtraction and addition (left to right):

$$4 - 3 = 1$$

$$1 + 1 = 2$$

Answer: 2

Mathematical Concept

Parentheses must be evaluated first in the order of operations. After parentheses are resolved, continue with the standard order: exponents, multiplication/division, addition/subtraction.

Common Mistakes to Avoid

- Performing operations without parentheses first: $24 \div 4 = 6$, then $6 + 2 = 8$, etc. (incorrect)
- Working strictly left to right without considering order of operations
- Dividing 24 by 4 only, ignoring the 2 inside the parentheses

Alternative Verification

Double-check each step: $(6 \text{ groups of } 4) \div 6 = 4$; $4 - 3 + 1 = 2$

Real-World Application

Recipe adjustments: If 24 tablespoons must be divided equally among $(4+2)$ servings, minus 3 tablespoons reserved, plus 1 extra, you use 2 tablespoons per serving.

2.2 Question 12: Converting Percentages (Discount)

Correct Answer: C

Problem

A \$50 shirt is on sale for 25% off. What is the final price?

Solution

Calculate the discount amount, then subtract from the original price.

1. **Step 1:** Calculate the discount:

$$\text{Discount} = 25\% \times \$50 = 0.25 \times \$50 = \$12.50$$

2. **Step 2:** Subtract from original price:

$$\text{Final Price} = \$50 - \$12.50 = \$37.50$$

Answer: \$37.50

Mathematical Concept

A percentage discount reduces the original price. The final price is the original price minus the discount amount, or equivalently, the original price times $(1 - \text{discount percentage})$.

Common Mistakes to Avoid

- Finding 25% of 50 correctly (\$12.50) but then adding instead of subtracting
- Calculating 25% as 0.025 instead of 0.25
- Forgetting to subtract the discount amount

Alternative Verification

Method 2: If 25% is discounted, you pay $100\% - 25\% = 75\%$ of the original price:

$$75\% \times \$50 = 0.75 \times \$50 = \$37.50 \quad \checkmark$$

Real-World Application

Shopping: A 25% discount on a \$50 shirt saves you \$12.50, making your purchase \$37.50.

2.3 Question 13: Ratio and Proportion

Correct Answer: C

Problem

A recipe calls for flour and sugar in a ratio of 3:2. If you use 8 cups of sugar, how much flour is needed?

Solution

Set up a proportion using the given ratio.

1. **Step 1:** Write the ratio of flour to sugar: $\frac{\text{flour}}{\text{sugar}} = \frac{3}{2}$

2. **Step 2:** Set up a proportion with the unknown:

$$\frac{3}{2} = \frac{x}{8}$$

where x is the amount of flour needed.

3. **Step 3:** Cross-multiply:

$$3 \times 8 = 2 \times x$$

4. **Step 4:** Solve for x :

$$\begin{aligned} 24 &= 2x \\ x &= \frac{24}{2} = 12 \end{aligned}$$

5. **Step 5:** Check: $\frac{12}{8} = \frac{3}{2}$

Answer: 12 cups of flour

Mathematical Concept

A ratio compares two quantities. A proportion states that two ratios are equal. When one part of a proportion changes, we can find the corresponding change in the other part.

Common Mistakes to Avoid

- Setting up the ratio backwards: $\frac{2}{3}$ instead of $\frac{3}{2}$ (incorrect)
- Solving $2x = 8$ instead of $2x = 24$
- Forgetting to cross-multiply correctly

Alternative Verification

Scaling factor: $\frac{8}{2} = 4$. So flour is $3 \times 4 = 12$ cups

Real-World Application

Cooking: Recipes use ratios to maintain proportions. A 3:2 ratio of flour to sugar scales consistently whether you double or halve the recipe.

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2.4 Question 14: Multi-Step Percentage

Correct Answer: C

Problem

A house costs \$200,000. It increases in value by 20%, then decreases by 10%. What is the final value?

Solution

Apply percentage changes sequentially.

1. **Step 1:** Calculate the 20% increase:

$$\text{Increase} = 20\% \times \$200,000 = 0.20 \times \$200,000 = \$40,000$$

2. **Step 2:** New value after increase:

$$\$200,000 + \$40,000 = \$240,000$$

3. **Step 3:** Calculate the 10% decrease on the new value:

$$\text{Decrease} = 10\% \times \$240,000 = 0.10 \times \$240,000 = \$24,000$$

4. **Step 4:** Final value after decrease:

$$\$240,000 - \$24,000 = \$216,000$$

Answer: \$216,000

Mathematical Concept

When applying consecutive percentage changes, each change is applied to the result of the previous step, not the original amount. Order matters in multiplication, but sequential percentage applications are sequential operations.

Common Mistakes to Avoid

- Adding 20% and subtracting 10% from the original: $200,000 + 40,000 - 20,000 = 220,000$ (incorrect; you'd be subtracting 10% of the original, not the new value)
- Calculating the 10% decrease on the original \$200,000 instead of \$240,000
- Assuming a 20% increase and 10% decrease nets to a 10% overall increase

Alternative Verification

Multiplier method: $200,000 \times 1.20 \times 0.90 = 200,000 \times 1.08 = 216,000$

Real-World Application

Real estate: Property values fluctuate. A house gaining 20% then losing 10% ends up ahead overall (8% net increase).

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2.5 Question 15: Fraction to Percent

Correct Answer: **D**

Problem

Convert to a percentage: $\frac{3}{5}$

Solution

Convert the fraction to a decimal, then multiply by 100%.

1. **Step 1:** Divide to get a decimal: $3 \div 5 = 0.6$
2. **Step 2:** Multiply by 100%:
 $0.6 \times 100\% = 60\%$

Answer: 60%

Mathematical Concept

To convert a fraction to a percentage, divide the numerator by the denominator to get a decimal, then multiply by 100% (or move the decimal point two places to the right).

Common Mistakes to Avoid

- Dividing the wrong way: $5 \div 3$ instead of $3 \div 5$ (incorrect)
- Multiplying by 100 without understanding it's converting to a percentage scale
- Writing the answer as 0.6% instead of 60%

Alternative Verification

Check: $60\% = \frac{60}{100} = \frac{3}{5}$

Real-World Application

Grades: If you answer 3 out of 5 questions correctly, you score 60%.

—

2.6 Question 16: Negative Numbers

Correct Answer: A

Problem

Subtract: $-3 - 5$

Solution

Subtracting is equivalent to adding the opposite.

1. **Step 1:** Rewrite subtraction as adding the opposite:

$$-3 - 5 = -3 + (-5)$$

2. **Step 2:** Add two negative numbers (move left on the number line):

$$-3 + (-5) = -(3 + 5) = -8$$

Answer: -8

Mathematical Concept

Subtracting a positive number is the same as adding a negative number. When adding two negative numbers, add their absolute values and apply the negative sign to the result.

Common Mistakes to Avoid

- Computing $-3 - 5 = 2$ (wrong sign)
- Thinking $-3 - 5 = -3 + 5 = 2$ (incorrect operation reversal)
- Losing track of which number is positive and which is negative

Alternative Verification

Number line: Start at -3 , move 5 units left (negative direction) to reach -8

Real-World Application

Bank accounts: Starting at $-\$3$ (in debt) and losing another $\$5$ results in $-\$8$ debt.

—

2.7 Question 17: Multiplying Negatives

Correct Answer: A

Problem

Multiply: $3 \times (-4)$

Solution

A positive number times a negative number yields a negative result.

1. **Step 1:** Recognize the signs: positive $\times$ negative
2. **Step 2:** Multiply the absolute values: $3 \times 4 = 12$
3. **Step 3:** Apply the sign: negative

$$3 \times (-4) = -12$$

Answer: -12

Mathematical Concept

Sign rules for multiplication:

- Positive $\times$ Positive = Positive
- Negative $\times$ Negative = Positive
- Positive $\times$ Negative = Negative
- Negative $\times$ Positive = Negative

Common Mistakes to Avoid

- Forgetting the negative sign and getting 12
- Mistakenly thinking positive $\times$ negative = positive
- Confusing multiplication with addition of negatives

Alternative Verification

Interpretation: Three groups of -4 items each: $(-4) + (-4) + (-4) = -12$

Real-World Application

Temperature: If the temperature drops 4 degrees per hour for 3 hours, the total change is $3 \times (-4) = -12$ degrees.

2.8 Question 18: Multiple Fractions

Correct Answer: C

Problem

Add: $\frac{1}{2} + \frac{1}{6}$

Solution

Find a common denominator and add.

1. **Step 1:** Find the LCD of 2 and 6:

- Multiples of 2: 2, 4, 6, 8, ...
- Multiples of 6: 6, 12, 18, ...
- LCD = 6

2. **Step 2:** Convert fractions to have denominator 6:

$$\frac{1}{2} = \frac{1 \times 3}{2 \times 3} = \frac{3}{6}$$
$$\frac{1}{6} = \frac{1}{6}$$

3. **Step 3:** Add:

$$\frac{3}{6} + \frac{1}{6} = \frac{4}{6} = \frac{2}{3}$$

Answer: $\frac{2}{3}$

Mathematical Concept

To add fractions with different denominators, find the LCD, convert each fraction, then add the numerators. Always simplify the final answer.

Common Mistakes to Avoid

- Using 12 as the LCD when 6 works (inefficient but not incorrect if properly simplified)
- Forgetting to simplify: leaving the answer as $\frac{4}{6}$ instead of $\frac{2}{3}$
- Adding denominators: $\frac{1}{2} + \frac{1}{6} = \frac{1}{8}$ (incorrect)

Alternative Verification

Convert to decimals: $\frac{1}{2} = 0.5$ and $\frac{1}{6} \approx 0.167$, so $0.5 + 0.167 \approx 0.667$. Check: $\frac{2}{3} \approx 0.667$

Real-World Application

Cooking: $\frac{1}{2}$ cup plus $\frac{1}{6}$ cup equals $\frac{2}{3}$ cup of an ingredient.

2.9 Question 19: Tax Calculation

Correct Answer: A

Problem

A book costs \$58.50. With 12% sales tax, what is the total cost?

Solution

Calculate the tax amount and add it to the original price.

1. **Step 1:** Calculate 12% tax:

$$\text{Tax} = 12\% \times \$58.50 = 0.12 \times \$58.50 = \$7.02$$

2. **Step 2:** Add tax to original price:

$$\text{Total} = \$58.50 + \$7.02 = \$65.52$$

Answer: \$65.52

Mathematical Concept

Sales tax is added to the base price. The total cost is the original price plus the tax amount, or equivalently, the original price times $(1 + \text{tax rate})$.

Common Mistakes to Avoid

- Calculating 12% of 58.50 incorrectly
- Subtracting tax instead of adding
- Forgetting to add the tax to the base price
- Rounding errors in decimal arithmetic

Alternative Verification

Method 2: $\$58.50 \times 1.12 = \65.52

Real-World Application

Shopping: When you buy a \$58.50 book, 12% sales tax adds \$7.02, making your total \$65.52.

2.10 Question 20: Comparing Forms

Correct Answer: A

Problem

Show that $\frac{3}{4} = 0.75 = 75\%$

Solution

Convert the fraction through each form to verify equivalence.

1. **Step 1:** Convert fraction to decimal:

$$\frac{3}{4} = 3 \div 4 = 0.75$$

2. **Step 2:** Convert decimal to percentage:

$$0.75 \times 100\% = 75\%$$

3. **Step 3:** Verify all three forms represent the same value:

$$\frac{3}{4} = 0.75 = 75\%$$

Answer: All three forms are equivalent

Mathematical Concept

Fractions, decimals, and percentages are different representations of the same quantity. Being fluent in converting between forms is essential for mathematical flexibility.

Common Mistakes to Avoid

- Miscalculating the decimal: $3 \div 4 \neq 0.34$ (incorrect)
- Forgetting to multiply by 100 when converting decimal to percentage
- Assuming fractions and percentages are always different representations

Alternative Verification

Work backwards: $75\% = \frac{75}{100} = \frac{3}{4}$ (after simplifying by dividing by 25)

Real-World Application

Grades: A score of $\frac{3}{4}$ on a test is the same as 0.75 or 75%, all expressing the same performance level.

3 Advanced Level: Complex Problem-Solving (Questions 21–30)

3.1 Question 21: Complex Fractions

Correct Answer: **B**

Problem

Add: $\frac{5}{8} + \frac{1}{6}$

Solution

Find the LCD and add the fractions.

1. **Step 1:** Find the LCD of 8 and 6:

- Prime factorization of 8: 2^3
- Prime factorization of 6: 2×3
- LCD = $2^3 \times 3 = 24$

2. **Step 2:** Convert to common denominator:

$$\begin{aligned}\frac{5}{8} &= \frac{5 \times 3}{8 \times 3} = \frac{15}{24} \\ \frac{1}{6} &= \frac{1 \times 4}{6 \times 4} = \frac{4}{24}\end{aligned}$$

3. **Step 3:** Add:

$$\frac{15}{24} + \frac{4}{24} = \frac{19}{24}$$

4. **Step 4:** Check if simplification is needed:

- Factors of 19: 1, 19 (prime)
- Factors of 24: 1, 2, 3, 4, 6, 8, 12, 24
- GCD = 1, so $\frac{19}{24}$ is in simplest form

Answer: $\frac{19}{24}$

Mathematical Concept

When denominators are coprime (share no common factors other than 1), the LCD is their product. When they share factors, use prime factorization to find the LCD efficiently.

Common Mistakes to Avoid

- Using 48 as the LCD instead of 24 (using the product instead of the true LCD)
- Miscalculating conversion factors
- Incorrectly identifying whether the result can be simplified

Alternative Verification

Decimal check: $\frac{5}{8} = 0.625$ and $\frac{1}{6} \approx 0.167$, so sum ≈ 0.792 . Check: $\frac{19}{24} \approx 0.792$

Real-World Application

Sewing: Combining $\frac{5}{8}$ yard of fabric with $\frac{1}{6}$ yard requires a total of $\frac{19}{24}$ yard.

—

3.2 Question 22: Exponents

Correct Answer: **B**

Problem

Evaluate: $5^2 + 4^2$

Solution

Calculate each square, then add.

1. **Step 1:** Calculate 5^2 :

$$5^2 = 5 \times 5 = 25$$

2. **Step 2:** Calculate 4^2 :

$$4^2 = 4 \times 4 = 16$$

3. **Step 3:** Add:

$$25 + 16 = 41$$

Answer: 41

Mathematical Concept

Exponents must be calculated before addition. $(5 + 4)^2 = 81$ is different from $5^2 + 4^2 = 41$. This demonstrates that exponents do not distribute over addition.

Common Mistakes to Avoid

- Adding first: $(5 + 4)^2 = 9^2 = 81$ (incorrect)
- Multiplying base and exponent: $5 \times 2 + 4 \times 2 = 18$ (incorrect)
- Computing $(5^2 + 4)^2$ or other order-of-operations errors

Alternative Verification

This is a Pythagorean triple component: $3^2 + 4^2 = 5^2$ (famous), but $5^2 + 4^2 = 25 + 16 = 41$

Real-World Application

Construction: If you have a 5-unit and 4-unit board, their combined square footage is $5^2 + 4^2 = 41$ square units.

—

3.3 Question 23: Square Roots

Correct Answer: C

Problem

A square room has an area of 169 square feet. What is the length of each side?

Solution

Use the formula for the area of a square and solve for the side length.

1. **Step 1:** Set up the equation:

$$\text{Area} = s^2 = 169$$

where s is the side length.

2. **Step 2:** Solve by taking the square root:

$$s = \sqrt{169}$$

3. **Step 3:** Find the square root (recognizing perfect squares):

- $12^2 = 144$ (too small)
- $13^2 = 169$ (correct!)

4. **Step 4:** State the answer:

$$s = 13 \text{ feet}$$

Answer: 13 feet

Mathematical Concept

For a square with area A , the side length is $s = \sqrt{A}$. Square roots are the inverse operation of squaring.

Common Mistakes to Avoid

- Dividing by 2: $169 \div 2 = 84.5$ (incorrect)
- Forgetting units in the answer
- Approximating when an exact answer exists

Alternative Verification

Check: $13 \times 13 = 169$

Real-World Application

Interior design: A square room with 169 square feet of floor space has sides of 13 feet each.
—

3.4 Question 24: Compound Percent

Correct Answer: D

Problem

An investment of \$100 increases by 8% in the first year and then by 7% in the second year. What is the final value?

Solution

Apply sequential percentage increases.

1. **Step 1:** Calculate value after first year (8% increase):

$$\text{Value}_1 = \$100 \times 1.08 = \$108$$

2. **Step 2:** Calculate value after second year (7% increase on the new value):

$$\text{Value}_2 = \$108 \times 1.07 = \$115.56$$

Answer: \$115.56

Mathematical Concept

Compound growth applies successive percentage increases to the accumulated amount. The formula is:

$$\text{Final Value} = \text{Initial} \times (1 + r_1) \times (1 + r_2) \times \dots$$

where r_i represents each rate of change.

Common Mistakes to Avoid

- Adding percentages: $\$100 + 8\% + 7\% = \115 (ignoring the time-based compounding)
- Calculating the 7% increase on the original \$100 instead of on \$108
- Misunderstanding that the second year's growth applies to the first year's result

Alternative Verification

Direct calculation: $100 \times 1.08 \times 1.07 = 100 \times 1.1556 = \115.56

Real-World Application

Investment growth: An initial investment of \$100 growing at 8% annually and then 7% reaches \$115.56 after two years.

3.5 Question 25: Advanced Order of Operations

Correct Answer: D

Problem

Simplify: $\left(\frac{1}{2} + \frac{1}{4}\right) \times 8 - 2$

Solution

Work through parentheses, then multiplication, then subtraction.

1. **Step 1:** Simplify inside parentheses. Find LCD of 2 and 4:

$$\begin{aligned} \frac{1}{2} &= \frac{2}{4} \\ \frac{2}{4} + \frac{1}{4} &= \frac{3}{4} \end{aligned}$$

2. **Step 2:** Rewrite: $\frac{3}{4} \times 8 - 2$

3. **Step 3:** Multiply:

$$\frac{3}{4} \times 8 = \frac{3 \times 8}{4} = \frac{24}{4} = 6$$

4. **Step 4:** Rewrite: $6 - 2$

5. **Step 5:** Subtract:

$$6 - 2 = 4$$

Answer: 4

Note: The provided answer key indicates the answer should be 10, but based on the problem statement and standard order of operations, the correct answer is 4. However, if the problem intended $(\frac{1}{2} + \frac{1}{4}) \times 8 + 2$, the answer would be $6 + 2 = 8$. If the problem is $(\frac{1}{2} + \frac{1}{4}) \times (8 - 2)$, the answer would be $\frac{3}{4} \times 6 = 4.5$. Given the provided answer is D = 10, there may be a transcription discrepancy. For pedagogical purposes, this solution demonstrates the correct application of order of operations.

Mathematical Concept

Parentheses create groupings that must be evaluated first. Only after parentheses are resolved do we proceed with multiplication and division (left to right), then addition and subtraction (left to right).

Common Mistakes to Avoid

- Multiplying before adding inside parentheses
- Forgetting the parentheses entirely
- Subtracting before multiplying

Alternative Verification

Step-by-step check: $(\frac{1}{2} + \frac{1}{4}) = \frac{3}{4}$; $\frac{3}{4} \times 8 = 6$; $6 - 2 = 4$

Real-World Application

Recipes with multiple operations: Combine ingredients (parentheses), multiply the resulting mixture by a scaling factor, then adjust the final amount.

3.6 Question 26: Complex Percentage

Correct Answer: C

Problem

An \$80 item is marked up by 50%, then discounted by 20%. What is the final price?

Solution

Apply the markup, then apply the discount to the marked-up price.

1. **Step 1:** Calculate 50% markup:

$$\text{Markup} = 50\% \times \$80 = 0.50 \times \$80 = \$40$$

2. **Step 2:** Price after markup:

$$\$80 + \$40 = \$120$$

3. **Step 3:** Calculate 20% discount on the marked-up price:

$$\text{Discount} = 20\% \times \$120 = 0.20 \times \$120 = \$24$$

4. **Step 4:** Final price after discount:

$$\$120 - \$24 = \$96$$

Answer: \$96

Mathematical Concept

Sequential operations must be applied in order. A markup increases the price first, then a discount applies to that new price. The combined effect is a net increase, since the 20% discount applies to the larger marked-up value.

Common Mistakes to Avoid

- Computing the discount on the original \$80: $20\% \times 80 = \$16$, giving $120 - 16 = \$104$ (incorrect)
- Adding and subtracting percentages: $+50\% - 20\% = +30\%$ applied to $\$80 = \104 (incorrect; doesn't account for sequential application)
- Reversing the order (discount then markup)

Alternative Verification

Multiplier method: $\$80 \times 1.50 \times 0.80 = \$80 \times 1.20 = \$96$

Real-World Application

Retail: A store marks up inventory 50%, then runs a 20% discount sale. Customers benefit, but the store still profits (12% net markup).

—

3.7 Question 27: Fraction/Decimal/Percent Equivalence

Correct Answer: **D**

Problem

Which is NOT equivalent to 0.2?

- A) $\frac{1}{5}$
- B) 20%
- C) 2%
- D) $\frac{1}{4}$

Solution

Convert each option to a common form (decimal) and compare.

1. **Step 1:** Convert option A: $\frac{1}{5} = 1 \div 5 = 0.2$ (equivalent)
2. **Step 2:** Convert option B: $20\% = \frac{20}{100} = 0.20 = 0.2$ (equivalent)
3. **Step 3:** Convert option C: $2\% = \frac{2}{100} = 0.02$ (NOT equivalent)
4. **Step 4:** Convert option D: $\frac{1}{4} = 1 \div 4 = 0.25$ (NOT equivalent)

Conclusion

Both C (2%) and D ($\frac{1}{4}$) are not equivalent to 0.2. The answer key specifies D as the answer; both would be correct.

Answer: D (though C is also not equivalent)

Mathematical Concept

Equivalent values must have the same decimal representation. $\frac{1}{5} = 0.2$ and $20\% = 0.2$, but $\frac{1}{4} = 0.25$ and $2\% = 0.02$, both different from 0.2.

Common Mistakes to Avoid

- Confusing 2% with 20%
- Miscalculating $\frac{1}{4} = 0.2$ when it actually equals 0.25
- Not converting to a common form before comparing

Alternative Verification

Work from 0.2: $0.2 = \frac{1}{5} = 20\%$. Compare: $\frac{1}{4} = 0.25 \neq 0.2$ and $2\% = 0.02 \neq 0.2$

Real-World Application

Data analysis: Distinguishing between equivalent representations (20% vs. 2%) is critical for correctly interpreting statistics and surveys.

3.8 Question 28: Negative Number Operations

Correct Answer: C

Problem

Evaluate: $(-3) \times (-2) + (-5)$

Solution

Apply multiplication first (order of operations), then addition.

1. **Step 1:** Identify the operations: multiplication and addition
2. **Step 2:** Perform multiplication first:

$$(-3) \times (-2) = 6 \quad (\text{negative times negative} = \text{positive})$$

3. **Step 3:** Rewrite: $6 + (-5)$

4. **Step 4:** Add:

$$6 + (-5) = 6 - 5 = 1$$

Answer: 1

Mathematical Concept

The sign rules for multiplication are critical:

- Negative $\times$ Negative = Positive
- Negative $\times$ Positive = Negative
- Positive $\times$ Negative = Negative
- Positive $\times$ Positive = Positive

After multiplication yields a positive result, adding a negative number is equivalent to subtraction.

Common Mistakes to Avoid

- Computing $(-3) \times (-2) = -6$ (incorrect sign)
- Forgetting to follow order of operations (multiplying before adding)
- Adding incorrectly: $6 + (-5) \neq 11$ or other errors

Alternative Verification

Rearrange: $(-3) \times (-2) + (-5) = 6 - 5 = 1$

Real-World Application

Accounting: Negative income (debt) times negative time (past) yields positive value; then subtract current expenses.

—

3.9 Question 29: Proportional Reasoning

Correct Answer: B

Problem

If 5 workers can complete a project in 12 days, how many days will it take 10 workers to complete the same project (assuming constant productivity)?

Solution

Use inverse proportional reasoning: more workers means fewer days needed.

1. **Step 1:** Recognize the inverse relationship:

$$\text{Workers} \times \text{Days} = \text{Constant (total work)}$$

2. **Step 2:** Calculate the constant:

$$5 \times 12 = 60 \text{ worker-days}$$

3. **Step 3:** Set up proportion with 10 workers:

$$10 \times d = 60$$

where d is the number of days.

4. **Step 4:** Solve for d :

$$d = \frac{60}{10} = 6 \text{ days}$$

Answer: 6 days

Mathematical Concept

Inverse proportionality states that as one variable increases, the other decreases such that their product remains constant. Doubling the workers halves the time needed.

Common Mistakes to Avoid

- Using direct proportion (doubling workers, doubling days): $10 \times 24 = 240$ (incorrect)
- Forgetting to account for the inverse relationship
- Arithmetic errors in calculating worker-days or final division

Alternative Verification

Logic check: 5 workers take 12 days; 10 workers (double) should take half the time: $12 \div 2 = 6$ days

Real-World Application

Project management: More team members working on a project reduces the timeline proportionally (assuming independent, non-overlapping tasks).

—

3.10 Question 30: Simple Interest

Correct Answer: B

Problem

You invest \$500 in a savings account with a 4% annual interest rate. How much will you have after one year?

Solution

Use the simple interest formula to find the interest earned, then add to principal.

1. **Step 1:** Identify the values:

- Principal (P) = \$500
- Rate (r) = 4% = 0.04
- Time (t) = 1 year

2. **Step 2:** Apply the simple interest formula:

$$I = P \times r \times t = \$500 \times 0.04 \times 1 = \$20$$

3. **Step 3:** Calculate total amount (principal + interest):

$$A = P + I = \$500 + \$20 = \$520$$

Answer: \$520

Mathematical Concept

Simple interest is calculated as a percentage of the principal applied only once (or once per specified period). The formula is:

$$I = P \times r \times t, \quad A = P + I = P(1 + rt)$$

This contrasts with compound interest, where interest is calculated on accumulated interest.

Common Mistakes to Avoid

- Calculating 4% of 500 as 0.04 instead of 0.04 (correct but easy to misunderstand)
- Forgetting to add interest back to principal
- Confusing simple interest with compound interest
- Miscalculating the percentage: $\$500 \times 0.4 = \200 (using 40% instead of 4%)

Alternative Verification

Percentage method: 4% of \$500 = $0.04 \times 500 = \$20$; total = $\$500 + \$20 = \$520$

Real-World Application

Banking: A \$500 deposit in a savings account at 4% annual interest earns \$20 in interest over one year, giving you a total of \$520.

Summary and Key Concepts

This answer key covers fundamental to advanced topics in arithmetic and financial mathematics:

1. **Order of Operations (PEMDAS):** Parentheses, Exponents, Multiplication/Division, Addition/Subtraction
2. **Fractions:** Adding, subtracting, multiplying, and dividing with common denominators
3. **Decimals:** Arithmetic operations and conversions to/from fractions and percentages
4. **Percentages:** Finding percentages, calculating discounts, markups, and tax
5. **Exponents and Roots:** Repeated multiplication and inverse operations
6. **Negative Numbers:** Operations with negative values and sign rules
7. **Ratios and Proportions:** Direct and inverse proportional relationships
8. **Financial Mathematics:** Percentages, taxes, discounts, simple interest, and investment growth

Mastery of these concepts builds a strong foundation for higher-level mathematics and practical problem-solving.